

TMS Automation – Productivity & Efficiency Tracking Requirement

1. Objective

The objective of this TMS enhancement is to automatically capture and analyze the end-to-end time spent by an associate on each transaction, across the client application, internal tools, Excel, portals, and other systems used to complete the transaction.

The solution should provide management with clear visibility into:

- * Actual productive time spent per transaction
- * Time spent across different applications/tools
- * Transaction processing time
- * Non-value-added activities
- * Idle/waiting time
- * Application switching time
- * Productivity per associate per day
- * Opportunities for automation and process improvement

The ultimate goal is to reduce non-value-added activities, improve associate productivity, increase process efficiency, and establish data-driven productivity benchmarks.

2. Business Scenario – AR Process Example

An associate logs into the system and works for approximately 8 hours per day.

For a typical AR transaction, the associate may perform the following activities:

1. Login to the client application.
2. Search and open the patient/account.
3. Review account details.
4. Review claim/payment/denial information.
5. Access payer portals or other external applications.
6. Perform the required AR action.

7. Update the outcome in the internal system.
8. Use Excel or other tools for additional tracking or calculations.
9. Complete the transaction.
10. Move to the next transaction.

For example:

Transaction 1

- * Client application review: 4 minutes
- * Payer portal: 2 minutes
- * Excel/internal tool: 2 minutes
- * Internal system update: 2 minutes
- * Total transaction time = 10 minutes

The TMS should automatically capture this transaction journey and calculate the actual time spent.

3. Required Automation

The TMS should automatically capture the associate's system activity during the working day without requiring manual time entry.

A. Login & Work Session Tracking

Capture:

- * Login time
- * Logout time
- * Total logged-in duration
- * Active working time
- * Idle/inactive time
- * Break time, where applicable
- * Productive time
- * Non-productive/non-value-added time

Example:

Metric Example

Login 8:00 AM

Logout 5:00 PM

Total logged-in time 9 hours

Breaks 1 hour

Available work time 8 hours

Active transaction time 6.5 hours

Idle/waiting time 45 mins

Non-value-added activity 45 mins

4. Transaction-Level Time Capture

Every transaction should have a unique Transaction ID and the TMS should capture the complete transaction lifecycle.

Required timestamps

- * Transaction started
- * Client application opened
- * Account accessed
- * Review started
- * External portal accessed
- * Internal tool accessed
- * Excel accessed
- * Action completed
- * Internal update completed
- * Transaction closed
- * Next transaction started

The system should calculate:

Total Transaction Processing Time = Transaction Completion Time – Transaction Start Time

Example:

Transaction	Start	End	Total Time
AR001	09:00:00	09:10:00	10 mins
AR002	09:10:30	09:18:30	8 mins
AR003	09:19:00	09:31:00	12 mins

5. Cross-Application Activity Tracking

One of the most important requirements is to identify where the associate spends time during transaction processing.

The TMS should capture time spent in:

- * Client software
- * Internal AR application
- * Payer portals
- * Excel
- * Email
- * Browser
- * Other approved business applications
- * Waiting/loading time
- * System errors/rework

Example:

Application	Time/Transaction	% of Transaction
Client application	4 mins	40%
Payer portal	2 mins	20%
Internal TMS	2 mins	20%
Excel	1 min	10%
Other	1 min	10%
Total	10 mins	100%

This will help identify where time is actually being consumed.

6. Productivity Calculation

The solution should automatically calculate productivity based on actual available working time and completed transactions, rather than simply measuring login hours.

Key metrics should include:

Transactions Per Hour

Transactions Completed ÷ Productive Hours

Example:

8-hour available workday

480 available minutes

10 minutes average transaction time

Expected capacity = $480 \div 10 = 48$ transactions/day

If the associate completes only 40 transactions, TMS should help identify where the remaining capacity was consumed.

7. Capacity & Productivity Opportunity

The system should compare:

Expected Productivity vs. Actual Productivity

Example:

Metric Result

Available minutes 480

Standard transaction time 10 mins

Expected transactions 48

Actual transactions 40

Productivity gap 8 transactions

Potential capacity gap 80 mins

The system should then identify why the 80 minutes were consumed.

For example:

- * 30 mins Excel activity
- * 20 mins waiting for client application
- * 15 mins duplicate data entry
- * 10 mins application switching

- * 5 mins other activity

This provides management with actionable information instead of simply reporting that productivity is low.

8. Non-Value-Added Activity Identification

The solution should identify activities that do not directly contribute to transaction completion.

Potential examples:

- * Repeated Excel updates
- * Duplicate data entry
- * Copy/paste between systems
- * Excessive application switching
- * Manual calculations
- * Searching for information across multiple systems
- * Waiting for applications/portals
- * Re-entering information already available in another system
- * Manual reporting/tracking
- * Rework caused by system/process limitations

These activities should be categorized and quantified.

9. Management Dashboard

The TMS should provide dashboards at:

Associate → Team → Process → Client → Project → Overall Operations

Daily dashboard

- * Associates logged in
- * Total transactions completed
- * Average transaction time
- * Productive time
- * Non-productive time
- * Idle time
- * Application-wise time
- * Productivity %
- * Expected vs. actual productivity
- * Productivity gap
- * Top non-value-added activities

Weekly/Monthly dashboard

- * Productivity trend
- * Transaction time trend
- * Capacity utilization
- * Process-wise efficiency
- * Application-wise time consumption
- * Automation opportunities

- * FTE/capacity opportunity

- * Improvement achieved

10. Management Insights

The system should move beyond reporting and provide actionable insights.

For example:

“Team A spends an average of 42 minutes per associate per day updating Excel trackers. This activity contributes approximately 850 hours of annual capacity consumption and is a potential automation opportunity.”

Another example:

“The average AR transaction time is 11.2 minutes. 2.1 minutes are spent navigating between the client application, payer portal and internal system. API/integration opportunity identified.”

Another:

“15% of available work time is consumed by non-value-added activities. Eliminating these activities could create additional transaction capacity without increasing headcount.”

11. Productivity Improvement Measurement

The TMS should establish a Before vs. After measurement.

Before automation

- * Average transaction time: 10 minutes
- * Transactions/day: 45
- * Excel activity: 45 mins/day
- * Application switching: 30 mins/day
- * Manual reporting: 20 mins/day

After automation

- * Average transaction time: 8 minutes
- * Transactions/day: 55
- * Excel activity: 10 mins/day
- * Application switching: 15 mins/day
- * Manual reporting: 5 mins/day

The platform should automatically calculate the improvement:

- * Transaction time reduction
- * Productivity improvement
- * Capacity released
- * Non-value-added time reduction
- * Potential FTE capacity
- * Financial benefit, where applicable

12. Key KPI Framework

The solution should provide the following KPIs:

- Average Transaction Processing Time
- Transactions Per Hour
- Transactions Per Day
- Productive Time %
- Non-Productive Time %
- Idle Time %
- Application Utilization
- Application Switching Time
- Excel Dependency
- Manual Activity Time
- Expected vs. Actual Productivity
- Capacity Utilization %
- Productivity Gap
- Non-Value-Added Time
- Automation Opportunity
- Capacity/FTE Opportunity
- Before vs. After Improvement

13. Important Design Principle

The objective is not to create a surveillance mechanism for associates.

The objective is to understand the process and technology inefficiencies affecting productivity.

The data should help answer:

“Where is the associate spending time, why is that time required, and what can we automate, eliminate, simplify or integrate?”

This will allow Operations, Business Excellence, Technology and Management teams to make decisions based on actual transaction-level data rather than assumptions or manually reported productivity.

14. Expected Business Benefits

Productivity

- * Increase transactions completed per associate
- * Improve utilization of available working hours
- * Establish data-driven productivity benchmarks

Efficiency

- * Reduce transaction processing time
- * Reduce application switching
- * Reduce duplicate data entry
- * Reduce Excel dependency

Automation

- * Identify high-volume manual activities
- * Prioritize automation opportunities
- * Identify API/integration opportunities

Management

- * Real-time productivity visibility

- * Accurate capacity planning
- * Data-driven staffing decisions
- * Transparent performance measurement

Financial

- * Release additional capacity
- * Reduce avoidable operational effort
- * Quantify automation benefits
- * Support FTE/capacity optimization

Expected Outcome

The TMS should ultimately answer five management questions:

1. How much time does an associate actually spend working?
2. How much time is spent completing each transaction?
3. Where is that time being spent across applications and activities?
4. How much of that time is non-value-added?
5. What can be automated or eliminated to improve productivity and capacity?

The expected outcome is a data-driven productivity and process intelligence platform that continuously identifies opportunities to reduce manual effort, improve transaction efficiency and increase operational capacity.